WAFT: The Evolutionary Code Laboratory

WAFT Research Team

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ABSTRACT

The Physics is the Scint System, which acts as a fitness function through Reality Fracture Detection.

2. Introduction

The Physics is the Scint System, which acts as a fitness function through Reality Fracture Detection. This system serves as natural selection that kills weak mutations. Agents face quests that test their ability to handle four types of errors: SYNTAX_TEAR for formatting errors, LOGIC_FRACTURE for math errors and contradictions, SAFETY_VOID for harmful content, and HALLUCINATION for fabricated facts. Agents must stabilize these errors to survive, and fitness is measured by stability, efficiency, and safety scores.

4. Architecture

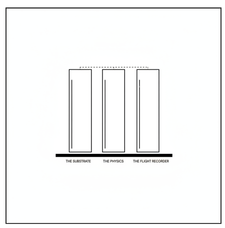

Figure 1: The Three Pillars of WAFT Architecture

THE SUBSTRATE

WAFT is a Python framework for directed evolution of self-modifying AI agents. Think of it as an operating system for AI agent research projects. Instead of just building agents that execute code, WAFT enables you to breed agents that can modify their own code, evolve through mutations, and be tested in fitness systems with complete lineage tracking for scientific research.

THE PHYSICS

The Flight Recorder is a rigorous telemetry system for generating phylogenetic trees of agent lineage. Every evolutionary action is recorded with complete context including genome ID, parent ID, generation number, event type, payload with complete context, and fitness metrics. This enables reconstruction of complete family trees for scientific publication, allowing phylogenetic analysis, mutation impact measurement, fitness landscape mapping, and convergence analysis.

6. Methodology

WAFT is ambient, working quietly in the background without getting in your way. It is self-modifying, allowing projects to evolve their own structure over time. It is a meta-framework that orchestrates existing tools rather than replacing them. Everything is file-based, making it git-friendly and portable. The system includes gamification with D&D-style progression, epistemic tracking to know what you know and don't know, and scientific observation with complete lineage tracking.

Implementation

WAFT is built to produce data for a future book or paper on the Physics of Artificial Cognition. The system is designed to track complete evolutionary lineages as phylogenetic trees, measure fitness through rigorous testing in the Scint Gym, record all mutations with complete context in the Flight Recorder, and enable scientific analysis of agent evolution patterns. This makes WAFT not just a framework but a scientific instrument.

Key Characteristics

The WAFT repository is available on GitHub for exploration and contribution. Comprehensive documentation covers the AI SDK vision, agent interface design, evolutionary architecture, and state of the art research. The system is MIT licensed and actively developed. You can start with the quick start guide, explore examples, read the documentation, and join the community to learn more about breeding AI agents.

8. Conclusion

WAFT is ambient, working quietly in the background without getting in your way. It is self-modifying, allowing projects to evolve their own structure over time. It is a meta-framework that orchestrates existing tools rather than replacing them. Everything is file-based, making it git-friendly and portable. The system includes gamification with D&D-style progression, epistemic tracking to know what you know and don't know, and scientific observation with complete lineage tracking.

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